

# KRA-DS-02: Synthesis of [RuCl<sub>2</sub>(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl<sub>2</sub>(CO)(p-cymene)] I

Date: 2025-02-20

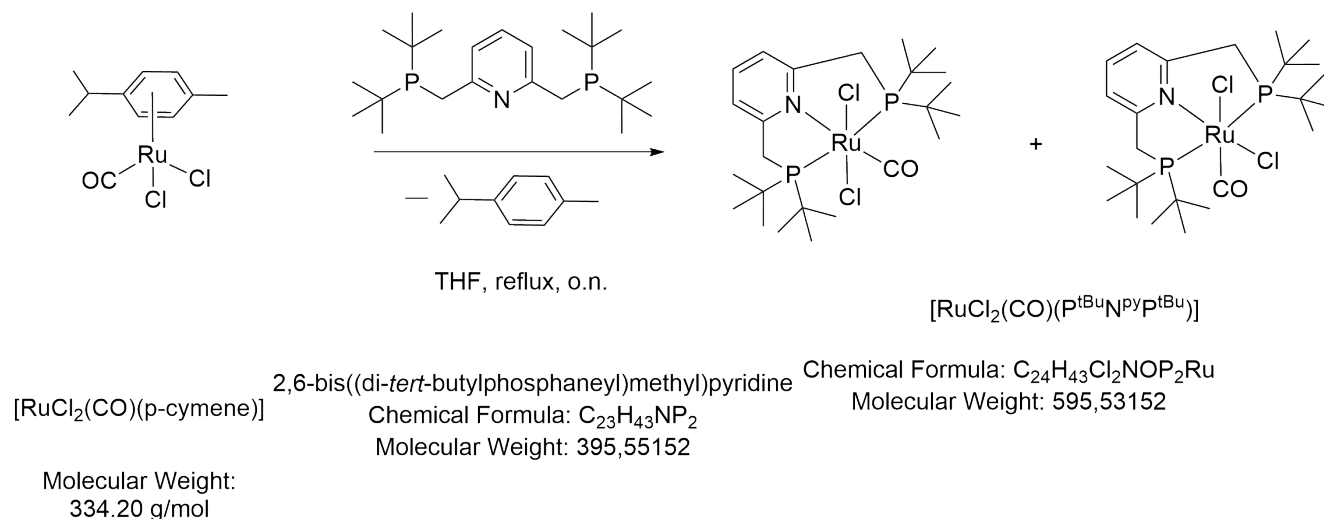
Tags: [RuCl<sub>2</sub>(CO)(p-cymene)] Dichloro PNP [RuCl<sub>2</sub>(CO)(PNP)] P(tBu)N(py)P(tBu) NMR 1H 31P KRA-DS

Category: Two-photon

Status: Done

Created by: David Schneider

## Reaction scheme/sample structure



## Literature/reference experiments

| Literature          | <a href="https://doi.org/10.1039/D1EE01053K">https://doi.org/10.1039/D1EE01053K</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reproduction        | /                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Related experiments | <p>Experiment - AE-157: Synthesis of [RuCl<sub>2</sub>(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl<sub>2</sub>(CO)(p-cymene)]</p> <p>Experiment - AE-230: Synthesis of [RuCl<sub>2</sub>(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl<sub>2</sub>(CO)(p-cymene)]</p> <p>Experiment - AE-232: Synthesis of [RuCl<sub>2</sub>(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl<sub>2</sub>(CO)(p-cymene)]</p> <p>Experiment - AE-261: Synthesis of [RuCl<sub>2</sub>(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl<sub>2</sub>(CO)(p-cymene)]</p> |

## Reagents

| Name                                                                                                                                          | Amount [mmol] | Equivalents | Purity | Mass <sub>theo,pure</sub> [mg] | Mass <sub>theo</sub> [mg] | Mass <sub>exp</sub> [mg] | Molar mass [g/mol] | Volume [ml] |
|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------|--------|--------------------------------|---------------------------|--------------------------|--------------------|-------------|
| [RuCl <sub>2</sub> (CO)(p-cymene)] ( Two-photon - AE-444: Synthesis of [RuCl <sub>2</sub> (CO)(p-cymene)] )                                   | 1.19          | 1.00        | 0.80   | 398.37                         | 497.96                    | 497.86                   | 334.20             | /           |
| PNP, 2,6-Bis(N,N-(tert-butyl)phosphinomethyl)pyridine ( Experiment - AE-253: Synthesis of 2,6-Bis(P,P-di(tertbutyl)phosphinomethyl)pyridine ) | 1.28          | 1.25        | 0.90   | 507.6                          | 564                       | 564                      | 395.55             | /           |
| THF ( Prep work - AE-397: Degassing of THF )                                                                                                  | /             | /           | /      | /                              | /                         | /                        | /                  | 50          |

## Work-up and Analytical Reagents

| Name      | CAS Number / Experiment Number             | Inventory number | Mass <sub>exp</sub> [g] | Volume [ml] | Concentration [M] |
|-----------|--------------------------------------------|------------------|-------------------------|-------------|-------------------|
| n-heptane | AE-315:<br>Degassing of n-Heptane          | /                | /                       | 30          | /                 |
| DCM-d2    | Prep work - AE-410:<br>Degassing of DCM-d2 | /                | /                       | 0.6         | /                 |

## Procedure/observations

All steps, unless mentioned otherwise, were carried out under argon using standard Schlenk technique or in an argon filled glovebox. (see Protocol [Protocol - Schlenk Technique](#))

| Date   | Time  | Step                                                                                                                                                          | Observations                                |
|--------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| 19.02. | 15:40 | A 100 mL and a 25 mL Schlenk flask was prepared                                                                                                               |                                             |
|        | 16:15 | An air condenser was set under vacuum for approx. one hour                                                                                                    |                                             |
|        | 16:00 | PNP(tBu) (564 mg) was weighted in the Glovebox K004 into the 100 mL Flask                                                                                     | white solid                                 |
|        | 16:20 | [RuCl <sub>2</sub> (CO)(p-cymene)] (597.86 mg) was weighted in the 25 ml Flask                                                                                | brick red powder                            |
|        | 17:30 | To the flask which contained the [Ru] 15 mL of THF were added                                                                                                 | dark red suspension; <a href="#">Ru.jpg</a> |
|        |       | To the flask containing the PNpTBu 15 mL of THF were added. The solution was stirred and shaken for 3 min.                                                    | suspension; <a href="#">PNpTBu.jpg</a>      |
|        |       | The PNpTBu solution was added to the reaction flask in one go                                                                                                 |                                             |
|        |       | The 25 mL flask was washed with 5 mL of THF because some solid (PNpTBu) stuck to the wall, The suspension was then also added to the reaction flask in one go |                                             |
|        |       | 15 mL of THF was added to the reaction.                                                                                                                       | dark red solution                           |
|        | 18:05 | The reaction mixture was stirred at rt and the flask was connected to the cooler in argon counterflow.                                                        |                                             |

|        |             |                                                                                                                                                                  |                                                                      |
|--------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
|        | 18:10-18:22 | The reaction mixture was heated to reflux                                                                                                                        | <a href="#">reflux.jpg</a>                                           |
|        | 18:22 -     | The reaction mixture was heated at reflux for approx. 13:40 h at 300 rpm.                                                                                        |                                                                      |
| 20.02. | - 10:02     | The reaction mixture was cooled down to r.t.                                                                                                                     | <a href="#">after_reflux.jpg</a><br>dark red suspension, precipitate |
|        | 10:50-11:35 | The volume of the remaining reaction mixture was reduced under reduced pressure, using external cooling trap to approx. 25 mL                                    | <a href="#">destillation.jpg</a><br><a href="#">reduced.jpg</a>      |
|        | 13:30       | To the solution n-Heptane (18 mL) was added under stirring.                                                                                                      | <a href="#">with_heptane.jpg</a><br>more precipitation               |
|        |             | The obtained mixture was filtered according to <a href="#">Protocol - Filtration with frit technique</a> and the residue was washed with n-Heptane (7 mL + 5 mL) | <a href="#">filtration.jpg</a>                                       |
|        | 15:00-16:55 | The obtained solid was dried under reduced pressure. <b>KRA-DS-02-1</b><br>Liquid phase: <b>KRA-DS-02-2</b>                                                      | <a href="#">dry_frit.jpg</a><br>yellow-brown solid                   |
| 25.02  | 15:00       | The solid was transferred into vial in a 10 mL Flask                                                                                                             | <a href="#">product.jpg</a>                                          |
|        | 15:45       | A NMR-sample was prepared from <b>KRA-DS-02-1</b> according to <a href="#">[Protocol]</a><br><a href="#">Preparation of NMR Sample in CD2Cl2.</a>                |                                                                      |

## Analysis

| Date   | Time           | Sample name | Analysis method                                     | Used device          | Solvent | Raw Data                                         | Processed Data                                              | Comparative data                                                                                                                                            | Interpretation                                                                                                                            |
|--------|----------------|-------------|-----------------------------------------------------|----------------------|---------|--------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 25.02. | 23:13<br>23:25 | KRA-DS-02-1 | <sup>1</sup> H,<br><sup>31</sup> P{ <sup>1</sup> H} | IAAC<br>400<br>MHz I | DCM-d2  | <a href="#">2025-02-26_13-01-44.zip</a>          | <a href="#">2025-02-26_13-01-44.zip_KRA-DS-02_10.nmrium</a> | Experiment - AE-230: Synthesis of [RuCl <sub>2</sub> (CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl <sub>2</sub> (CO)(p-cymene)] -2 | Desired product at 65 ppm (31P). All signals in <sup>1</sup> H found. Purity according to <sup>31</sup> P and <sup>1</sup> H of over 95 % |
| 07.04. | /              | KRA-DS-02-K | XRD                                                 | /                    | /       | <a href="#">Kristallstruktur-KRA-DS-02-K.zip</a> | /                                                           | /                                                                                                                                                           | 60% cis, 15% trans, 25% Ru(III)                                                                                                           |

## Product characterization

|             | mass [mg] | purity [%] | mass <sub>pure</sub> [mg] | amount [mmol] | Yield [%] |                                                 |
|-------------|-----------|------------|---------------------------|---------------|-----------|-------------------------------------------------|
| KRA-DS-02-1 | 407.63    | >95 %      | 407.63                    | 0.68          | 58        | yellow-brown solid, <a href="#">product.jpg</a> |

## Linked experiments

- JSC-KS-04: Degassing of n-Heptane
- AEI-047: Synthesis of 2,6-Bis(P,P-di(tertbutyl)phosphinomethyl)pyridine
- AEI-057: Synthesis of [RuCl<sub>2</sub>(CO)(p-cymene)]
- AEI-060: Synthesis of [RuCl<sub>2</sub>(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl<sub>2</sub>(CO)(p-cymene)]
- AEI-072: Synthesis of [RuCl<sub>2</sub>(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl<sub>2</sub>(CO)(p-cymene)]
- AE-100: Synthesis of [RuCl<sub>2</sub>(CO)(p-cymene)]
- AE-101: Synthesis of [RuCl<sub>2</sub>(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl<sub>2</sub>(CO)(p-cymene)]
- AE-102: Degassing of THF
- AE-108: Synthesis of [RuCl<sub>2</sub>(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl<sub>2</sub>(CO)(p-cymene)]
- AE-120: Degassing of n-Heptane
- AE-157: Synthesis of [RuCl<sub>2</sub>(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl<sub>2</sub>(CO)(p-cymene)]
- KRA-080: Degassing of THF
- AE-221: Synthesis of [RuCl<sub>2</sub>(CO)(p-cymene)]
- AE-230: Synthesis of [RuCl<sub>2</sub>(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl<sub>2</sub>(CO)(p-cymene)]
- AE-232: Synthesis of [RuCl<sub>2</sub>(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl<sub>2</sub>(CO)(p-cymene)]
- AE-248: Synthesis of [RuCl<sub>2</sub>(CO)(p-cymene)]
- AE-253: Synthesis of 2,6-Bis(P,P-di(tertbutyl)phosphinomethyl)pyridine
- AE-261: Synthesis of [RuCl<sub>2</sub>(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl<sub>2</sub>(CO)(p-cymene)]
- AE-315: Degassing of n-Heptane

Prep work - [AE-397: Degassing of THF](#)

Prep work - [AE-410: Degassing of DCM-d2](#)

Two-photon - [AE-444: Synthesis of \[RuCl2\(CO\)\(p-cymene\)\]](#)

## Linked resources

Protocol - [Preparation of NMR Sample](#)

Protocol - [Filtration with frit technique](#)

Protocol - [Schlenk Technique](#)

## Attached files

Kristallstruktur-KRA-DS-02-K.zip

sha256: 0d7f01d6b3a5251b8aee1ba4638e3b9bb7aca9ab53d5efc8c1764deb1d883fa4

2025-02-26\_13-01-44.zip\_KRA-DS-02\_10.nmrium

sha256: 6da7c3c6dc041caeea05a4222d1b563567f966f191c7928d394d55e686a3e812

2025-02-26\_13-01-44.zip

sha256: 69da50efdf66407a6f4c37364124c8398d8a43821b7d910dc0e78b28488f914d

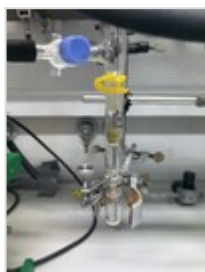
product.jpg

sha256: 041f7855fbc8094ad9a31d1acb5719a8f0667d16f2899581a4914cc2f5caa6fc



dry\_frit.jpg

sha256: c00f27e1abfdb22213de1f56b2f0437d09b14e21b260843f7db6990041a55914



filtration.jpg

sha256: bc177e952e308f238ed5f0e03d39185691d9a59ff5b335d922c016fc2bba895e



with\_heptane.jpg

sha256: b4df4383070e09d5d7a20629021a5999835dc4e38ba442a1497df0fd7d68aa94



reduced.jpg

sha256: c0c0c371d3611372f0e3126c1c77af3ff21fd69715f2cee70c480b1778b5605f



distillation.jpg

sha256: 3b6ad7c0d1489a07ceb06781b0be4ee40fe033986d74f730c10d9284051bc07b



after\_reflux.jpg

sha256: d49a2e84daa1b12c4d15823a35e002ffa14b77e27dff33c544589d4aac41d853



reflux.jpg

sha256: c949a680b61df2c28c92b352f2127a5f27944e81cd2d26c84e6727ca719ce593



PNPtBu.jpg

sha256: f09bb6dc336f25bcbc092d4f633794228f3e7ba1fb51211dad5d36cf8f080a95



Ru.jpg

sha256: 4c1c3f29051ae738359754460f44a2972e8b68d24132a8cab92c716210966358



## Comment

On 2025-02-24 09:39:47 Kristína Rabatinová wrote:

Notes:

- \* adding the analytical reagents table
- \* 19.02., 15:40, "A 100 ml and a 25 ml Schlenk flask were prepared." should be grey
- \* 19.02. How was the PNPTBu solution added to the reaction flask? --> adding dropwise or at once, adding under stirring ALSO here changing solution to suspension
- \* "The 25 ml flask was washed with 5 ml of THF, ...." --> I would add that this was done because some solid (PNPTBu) stuck to the flask
- \* "15 ml of THF was ..." --> was it a dark red solution or a suspension?
- \* 19.02. 18:22 - "the reaction mixture was heated AT reflux ..." since heated TO reflux is above
- \* again generally add more descriptions to the picture, are a "rule" every picture should have a description, How does the solution changes after reflux? Darker/brighter or not really? Is it a solution or a suspension? Do we see a precipitate? What happens when we add heptane? More precipitation? Color of the precipitation and the solution? How does the solid looks like on the frit? Color?



Unique eLabID: 20250220-68b2515edd676728dafdc65ef3c068d01953260f  
Link: <https://elab.water-splitting.org/experiments.php?mode=view&id=1776>